

Geophysical Workflow Report

Generated by PyHydroGeophysX Multi-Agent System

Executive Summary

****Workflow Execution Date:**** 2026-01-19 14:34:10

Original User Request:

Run a standard ERT inversion using the DAS-1 instrument.

Data file: 20171105_1418.Data

Electrode file: electrodes.dat

Workflow Configuration:

- Data File:
C:\Users\hchen117\AppData\Local\Temp\phgx_uploads_zf_gafmm\20171105_1418.Data
- Instrument: DAS-1
- Seismic Integration: No

Key Results:

- Loaded 56 electrodes with 930 measurements
- Inversion converged in 4 iterations (chi2: 0.762)
- Mean water content: 0.179

Narrative Summary

Executive Summary

The objective of the conducted workflow was to perform a standard Electrical Resistivity Tomography (ERT) inversion using data collected with the DAS-1 instrument. The specific data file analyzed was recorded on November 5, 2017, and encompassed a total of 56 electrodes and 930 measurements. The inversion process successfully converged in four iterations, yielding a final chi-squared value of 0.762, which indicates a good fit between the observed and modeled data. An additional aspect of the analysis involved assessing the mean water content, which was determined to be 0.179, with a range from 0.055 to 0.307.

Despite the successful execution of the ERT inversion, the integration of climate data into this workflow was not performed. This omission limits the ability to contextualize the resistivity changes observed in the ERT results with corresponding climatic variations such as precipitation, potential evapotranspiration (PET), and temperature. Typically, the resistivity of subsurface materials is influenced by water content, which is directly affected by climatic conditions. Therefore, understanding how recent weather patterns, particularly rainfall events and subsequent drying periods, impact soil moisture would provide valuable insights into the observed resistivity characteristics.

In the context of resistivity changes, it is important to note that increased rainfall typically leads to enhanced water content in the subsurface, resulting in lower resistivity values. Conversely, during periods of drying, the reduction in water content can lead to higher resistivity readings. The mean water content of 0.179 suggests that the subsurface retains a moderate level of moisture, likely influenced by recent climatic conditions. However, without specific climate data for the period surrounding the measurement date, it remains challenging to draw definitive conclusions regarding the relationship between resistivity and hydrological dynamics.

Key findings from the analysis highlight a moderate mean water content and a well-converged inversion; however, the absence of climate data raises several caveats regarding the interpretation of resistivity values. Future studies should prioritize the integration of climate data alongside ERT measurements to enhance the understanding of subsurface hydrology. Additionally, it is recommended to investigate seasonal variations and their impact on resistivity, which may require repeated ERT surveys and corresponding climate monitoring. This comprehensive approach will facilitate a more robust analysis of the interactions between climatic factors and subsurface resistivity, ultimately leading to improved hydrological modeling and resource management.

Data Processing Summary

ERT Data Loading

- Number of electrodes: 56
- Number of measurements: 930
- Quality metrics: N/A

****Insights:**** N/A

Climate Data Integration

No climate data was integrated in this workflow.

Cross-Modal Climate-ERT Analysis

Climate data not available for cross-modal analysis.

Inversion Results

ERT Inversion

- Final chi2: 0.7615702932153683
- Iterations: 4
- Convergence: Success

****Interpretation:**** N/A

Water Content Analysis

Water Content Statistics

- Mean water content: 0.179
- Range: [0.055, 0.307]

Uncertainty Analysis

- Mean uncertainty (σ): 0.037
- Maximum uncertainty: 0.063
- Number of realizations: 100

****Interpretation:**** N/A

Visualizations

Resistivity

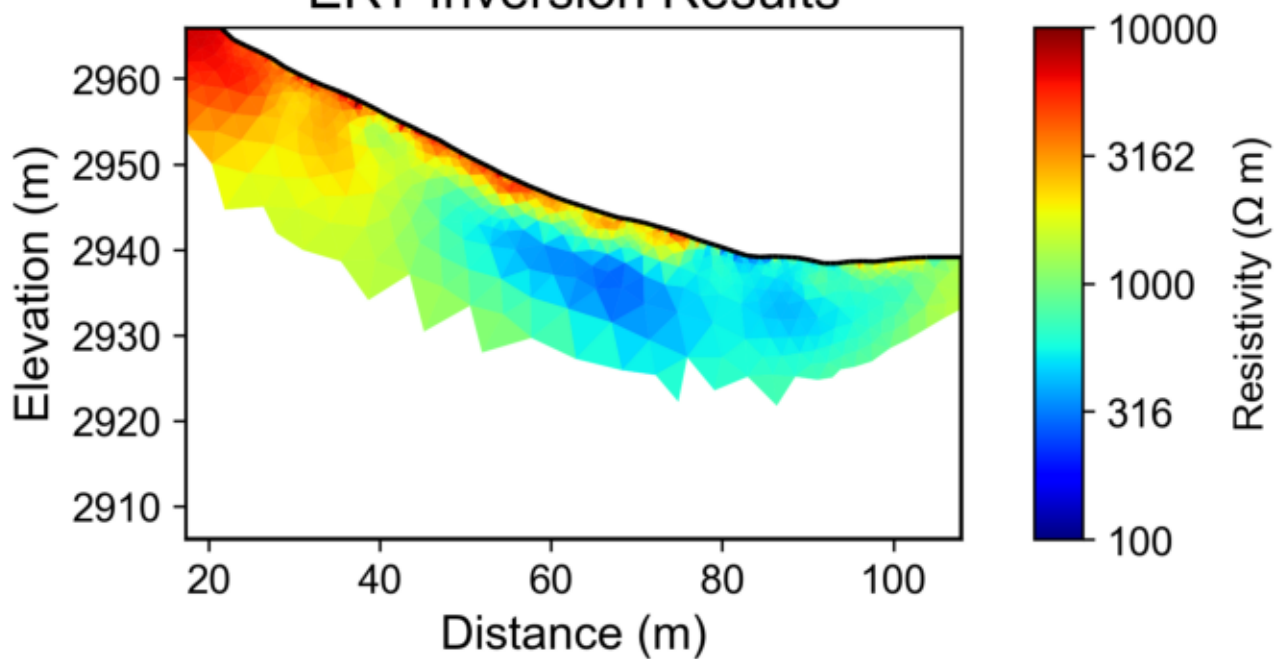
Water Content

Water Content Uncertainty

Report generated on 2026-01-19 14:34:22

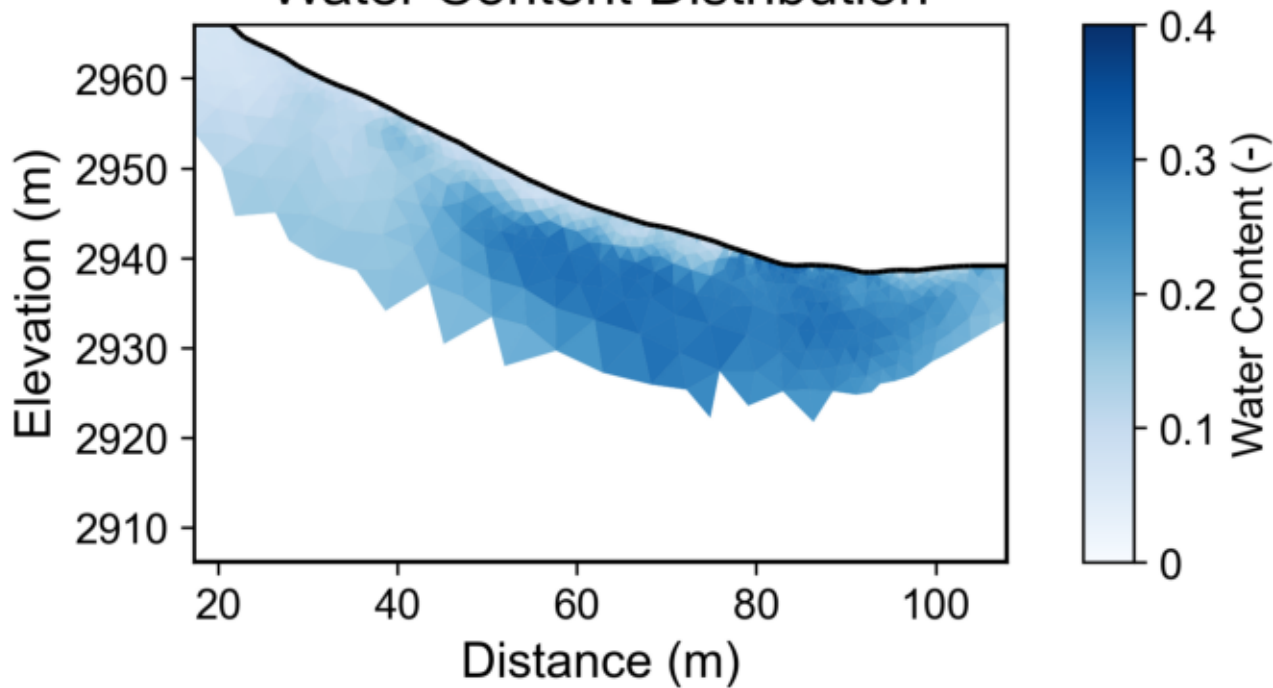
Resistivity

ERT Inversion Results



Water Content

Water Content Distribution



Water Content Uncertainty

Water Content Uncertainty

